# Thermal effects and energy budgets

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Estimate a temperature change only after stating how much energy enters the material.

**SUCCESS CRITERIA - I can...**

1. Apply an explicit absorbed-energy fraction.
2. Use mass and specific heat in a temperature estimate.
3. State the no-phase-change and no-loss assumptions.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

**LANGUAGE GOAL** Use “absorbed” and “specific heat” to explain your model or result. Sentence starters are optional.

**absorbed**: taken into the specified object or system

**specific heat**: energy per mass needed for a one-degree temperature change

**energy budget**: an account of all energy stored and transferred

## 01 Bring back an earlier idea

Bodies initially at 330 K and 350 K exchange heat in an isolated system.
**Explain** what happens to total energy and to the temperature-difference opportunity for work at equilibrium.

**Word help:** heat = energy transferred because of a temperature difference

**Sentence starter:** Both … . They differ in … because … .

**Marzano check - circle:** ① ② ③

## 02 Explain a worked example

**Reference:** $Q_{absorbed} = mc\Delta T$; $\Delta T = Q_{absorbed}/(mc)$.
First identify the absorbed energy. Use only when there is no phase change and c is approximately constant.

**Worked reasoning:** A surface receives 500 J. A model accounts for 200 J reflected, 250 J absorbed as heating and 50 J transmitted. The full account sums to 500 J.

**Your turn: Draw** proportional energy arrows and identify which part can directly raise this surface's temperature.

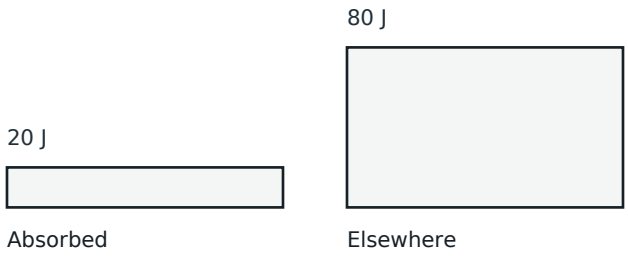
**Sentence starter:** The … in my model represents … . Its label shows … .

**Marzano check - circle:** ① ② ③

## 03 Represent and interpret

A model releases 100 J. Only 20% enters a sample.
**Draw** the energy partition and calculate energy absorbed.

**Sentence starter:** I found … [unit]. This value means … in this situation.




Bar height represents energy. Read the value and unit above each bar.

**Marzano check - circle:** ① ② ③

## 04 Solve and justify

A hypothetical energy release is $2\times10^6$ J. A .50 kg sample absorbs 1%; c=800 J/(kg·°C). Estimate$\Delta T$ without phase change or losses.

**Sentence starter:** I found … [unit]. This value means … in this situation.

**Marzano check - circle:** ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

## 05 Analyze the reasoning

A temperature prediction exceeds a material’s melting point, but the calculation used only mcΔT. What must be added or reconsidered?

**Sentence starter:** I predict … . My model supports this because … .

**Marzano check - circle:** 1 2 3

## 06 Choose a method

A 4 kg cart descends 6 m from rest.
**Use** g=10; losses are 8 J.
**Find** bottom speed.

**Sentence starter:** I found … [unit]. This value means … in this situation.

**Marzano check - circle:** 1 2 3

## 07 Exit check

A 2 kg sample with c=500 warms 8°C.
**Find** absorbed energy. If this is 4% of source energy, find source energy.

**Sentence starter:** I found … [unit]. This value means … in this situation.

**Marzano check - circle:** 1 2 3

## 08 STRETCH: Apply it another way

A 0.4 kg panel absorbs 1200 J, emits 300 J and does no work. With c=450 J/(kg·K), find internal-energy and temperature changes.

**Sentence starter:** I found … [unit]. This value means … in this situation.

**Marzano check - circle:** 1 2 3